XIAO PING SONG, PhD

Hands-on technical leader bridging consulting, delivery, and engineering to redesign enterprise workflows with platforms, data and AI. Across government, defence and critical infrastructure, I align stakeholders around measurable outcomes and lead teams to deliver with speed and clarity. I care deeply about team culture, transparency, and building systems that last.

Certified PMP, ITIL, and AWS Solutions Architect. Former Postdoctoral Fellow leading research and teaching in machine learning and geospatial programming.

EMPLOYMENT

2025 | Present

Manager, Data Engineering (Delivery & Solutions)
Esri Singapore

- Lead a team delivering ArcGIS Enterprise in secure, air-gapped environments, shaping engagements from technical discovery through proof of value to production rollout
- Establish and lead a portfolio of governed AI programmes spanning local inference, geospatial agents, presales qualification, and managed-service triage, with evaluation frameworks and human-review gates to validate reliability before production use
- Scale team capability through onboarding frameworks, delivery playbooks, shared agent skills, and peer-feedback systems

2024 | 2025

Senior Data Engineer, Tech Lead
Esri Singapore

- Managed technical rollout of 1000+ web widgets, geo-data layers and API endpoints to 1,200 users and 9 external systems, with all phases delivered on schedule
- Led cross-team initiatives and client negotiations, introducing change controls that reduced scope by 375 man-days
- Forward-deployed development for data migration, achieving 100% validation and sign-off

2021 | 2023

Postdoctoral Fellow, Project Manager, Adjunct Lecturer
NUS College of Design and Engineering

- Secured 3-year grant for $1.5 mil
- Assembled a cross-organisational team to develop sustainability reporting standards across six work streams
- Led deployment of IoT sensor networks, reducing costs by 60% through sharper requirements and fit-for-purpose wireless technologies

2020 | 2022

Postdoctoral Fellow, Project Manager
National University of Singapore, Faculty of Science

- Led the delivery of a biodiversity benchmarking product, securing senior-management buy-in
- Secured access to restricted high-res LiDAR and $60k for satellite images

2016 | 2020

PhD Researcher
ETH Zürich, SEC, Future Cities Laboratory

- Published four papers in leading global journals and presented research across the UK, US, Switzerland, and Japan

CONTACT

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CODE & PLATFORMS

Python / R / SQL / Git
LLM Apps / Agents / RAG
ArcGIS Enterprise / Online / Pro
AWS / Google Cloud / Azure
APIs (REST, GraphQL, MCP)
CI/CD (Azure DevOps, GitHub)
Containers / Orchestration / IaC (Docker / Kubernetes / Terraform)
Bash / PowerShell
QGIS / Google Earth Engine
Jira / Confluence / Linear / Notion

DOMAINS & METHODS

Technical Consulting / Presales
Enterprise Platform Delivery
AI Harnesses / Evals / Governance
Data Architecture / Migration
ML / Feature Engineering
GIS / Geospatial Analytics
Remote Sensing / Image Analysis
Utilities / IoT / Wireless Systems
Resource & Capacity Planning
Scrum / Agile Transformation
Curriculum Development

EDUCATION

2020 — **PhD in Architecture (Urban Planning / Analytics)**
National University of Singapore, College of Design and Engineering

- Publications at scholar.google.com

2016 — **B.Sc. (Hons) in Life Sciences,**
Minor in Technology Entrepreneurship
National University of Singapore

TEACHING & TRAINING

2023 — **Geospatial Data Science with R**
Udemy online course (600+ learners; rated 4.7 / 5)
bit.ly/geospatial-datascience-r_udemy

2023 — **Geodesign**
Module Lecturer for master's and bachelor's programmes in Landscape Architecture, National University of Singapore

2018 | 2022 — **Coding workshops: Spatial Statistics and Version Control**
Online and in-person workshops on R programming and Git
xpsong.com/posts/intro2r
xpsong.com/posts/intro2r-spatial

2016 | 2020 — **Weekly Music Classes**
Part-time Educator, My Drum School Pte Ltd. Attained Net Promoter Score of 75 (top 5%); Best Educator of the Quarter twice in 3 years
instagram.com/p/BzpCir0HQA-

ACCOMPLISHMENTS

Certifications

- ITIL® 4 Foundation (PeopleCert)
- AWS Certified Solutions Architect (AWS)
- Project Management Professional (Project Management Institute)
- Data Engineering Professional Certificate (DeepLearning.AI, AWS)
- MLOps Specialization (DeepLearning.AI)
- Deep Learning Specialization (DeepLearning.AI)
- Machine Learning Specialization (Stanford University)

Awards

- Most Valuable Player (Leadership), Esri Singapore (2024)
- SEC PhD Scholarship, National Research Foundation (2016)
- Best Full-time National Serviceman, Officer Cadet School (Midshipmen Wing), Singapore Armed Forces (2011)
- Jacob Ballas Scholar's Award, Perfect score (six distinctions) for GCE 'A' Level Examinations (2009)

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